

Quick Start Guide

This guide covers the shortest path from a fresh clone to the current browser dashboard.

Prerequisites

Tool	Why	Install
Miniconda or Anaconda	Python environment manager for the workflow	Download installer
Node.js 20 LTS or newer	Runs the Next.js vizualizer	Download installer or use <code>winget install OpenJS.NodeJS.LTS</code>

1. Create the Conda Environment

From the repo root:

```
conda env create -f envs/environment.yaml -n pypsa
conda activate pypsa
```

The environment file installs Snakemake, Python packages, and the solver stack used by the workflow.

If Python 3.13 is selected, apply the current `google-cloud-storage` fix after creating the environment:

```
conda run -n pypsa pip install "google-cloud-storage>=2.10"
```

2. Download Required Data

The repo does not ship the large `data/` payload. Download the cutout and Zenodo-backed datasets before running scenarios:

```
cd personal_data_download
python download_cutout.py
python download_zenodo_files.py
```

3. Install the Vizualizer Dependencies

```
cd vizualizer  
npm install
```

4. Start the Web Dashboard

```
cd vizualizer  
npm run dev
```

Open `http://localhost:3000` in the browser.

If the server does not pick up the intended conda environment automatically, set `PLANUI_CONDA_ENV=pypsa` before starting `npm run dev`.

5. First Things to Verify

- The Scenario Builder loads a YAML template and allows edits through the form controls.
 - The Run Queue tab can list and tail jobs.
 - The Results tab shows only folders with the required comparison CSVs.
-

Optional Production Start

```
cd vizualizer  
npm run build  
npm start
```